Manufacturing (Food & Beverage)

Project #	Industry	Project Title (students)	Project Description (students)
1	Manufacturing (Food & Beverage)	Brand Power Model	In this Capstone project, you will create a Brand Power Index for a Forbes Global 2000 multinational beverage company. The sponsor runs an ongoing consumer research program in which it conducts consumer surveys to understand consumers' perceptions and feelings towards its beverages. The key outputs of the survey are brand centric metrics that generate a "Brand Power Score" that represents the overall strength of a brand relative to its competitors. Based on the data provided, you will develop a model to determine a Brand Power Index and its three drivers. Focusing on one (1) brand, you will analyze its performance and develop a recommended marketing strategy for it to maximize its Brand Power growth. **Requirements: R or Python, SQL, Data visualization** **Preferred skills: Familiarity with or interest in Consumer Insights & Brand Strategy** **Internal data**
2	Manufacturing (Construction Equipment)	Shipping route optimization	In this project, you will build a model and application to optimize shipping routes for a leading manufacturer of agricultural and construction equipment. This project aims to develop a model and application that can process and analyze shipping data; calculate distances and travel times between locations; optimize shipping routes based on defined criteria; identify the best multi-stop routes, ensuring compliance with equipment and capacity constraints; and provide interactive visualizations and

			detailed reports for user decision-making. This project offers a valuable opportunity to tackle a real-world logistics challenge for a large industry. **Requirements: R** **Internal data**
3	Manufacturing (Awards & Gifts)	Building a predictive model of the existing sales funnel	In this Capstone project, you will help a manufacturer of crystal awards and personalized gifts build a predictive model of the company's existing sales funnel. This model should be broken out by each sales territory and each salesperson. Using historical data, you will project the number of new customers and new projects expected in 2025 and the average value of revenue expected for these. Your model should also project retention from the previous year and break retention into quartiles (e.g. projects between $1,000-3,000 retain at 80%, projects $3,000-6,000 retain at 90% etc.). There is a vast amount of historical data from the last several years that the sponsor will make available for you to identify trends and patterns across customers, products, and communications. The sponsor is very eager to hear your recommendations on how it can prioritize its ideal customers and improve its sales process, customer communication, and product mix. **Requirements: R, Excel, Predictive modeling** **Internal data**
4	Transportation & Logistics	Planning a supply chain for a new customer	In this Capstone project, you will work with a global logistics company to plan a supply chain for a new customer. You will be given an internal training dataset used to onboard new employees of the sponsor. Your task will be to research and create a robust network design model with logical assumptions, parameters and trade-offs. Your final report will use sensitivity analysis and what if scenarios to present recommendations to the sponsor's management team. **Requirements: Python (e.g. PuLP library)**

			Preferred skills: Excel Solver or other network design software **Internal training data**
5	Food & Beverage Services	Optimizing quick service restaurant pricing: Managing profit and consumer expectations	In this Capstone project, you will work with a parent company of multiple foodservice franchises to determine optimal pricing for the most popular product at three of the sponsor's restaurant brands respectively. Optimal pricing is based on many factors including geography, competitive pressure, price elasticity of demand, discount elasticity of demand, and company value proposition. Your analysis of optimal pricing will contribute significantly to the long-term profitability of three of the sponsor's brands. **Requirements: Python, Elasticity of demand modelling** **Internal data**
6	Consulting (Mergers & Acquisitions)	Predictive modeling for identifying companies likely to sell	In this Capstone project, you will work with a leading Consulting and Mergers & Acquisitions (MA) advisory firm to build a model that predicts companies likely to sell their business in the near future. Specifically, you will explore whether publicly available data, SEC Investment Adviser Public Disclosure (IAPD) data (which provide information on financial advisor businesses) and relevant economic data/indicators, can serve as predictive signals for companies likely to sell. In addition to a Predictive Scoring Model that assigns a “likelihood to sell” score to companies based on identified data patterns, you will establish a streamlined process for the sponsor's associates to ingest and refine raw SEC Investment Adviser Public Disclosure (IAPD) data; apply the predictive model to an analysis layer to generate actionable insights; and output a ranked list of companies with associated sale probability scores. You will have the opportunity to explain your model selection process (e.g., linear regression, decision tree, neural network etc.) and share your insights and lessons learned during model development. Your work will be a valuable contribution to the sponsor's strategic initiatives to enhance its market intelligence and proactively engage high-potential sellers.

			Requirements: R or Python/Pyspark (Python preferred), Machine learning **Internal and public data**

7	Retail (Furniture)	Activating customer experience platform with predictive models	In this project, you will build a predictive model for an established home furnishings retailer. The sponsor uses the One Voice platform to gather customer feedback on their experience at different touchpoints. This project aims to focus on the in-store and online Customer Effort Score (CES), which measures the ease of doing business with the company. The goal is to build a predictive model that estimates how low CES scores impact sales and identify areas of improvement from low CES score data. Additionally, the project will explore the relationship between in-store and online CES scores. The primary project objectives are: 1.) Develop a predictive model to estimate the impact of low CES scores on sales. 2.) Identify opportunities for improvement based on low CES score data. 3.) Understand the correlation between in-store and online CES scores and their impact on each other. The model you build will help to optimize the sponsor's business decisions. **Requirements: Python, SQL (e.g. star schema data model), Tableau** **Internal data**
8	Transportation & Logistics	Predicting bus passenger counts and understanding how vehicular traffic is affected by	In this Capstone project, you will work with a large regional transportation agency to analyze data that will help them improve their infrastructure and network. Two areas the sponsor would like to focus on are: 1. Predicting passenger counts and bus needs at their main transit hub.

		seasonality, weather, holidays, and other events	2. Understanding how bridge and tunnel traffic is affected by seasonality, weather, holidays and other events. By focusing on these two areas, you will uncover patterns in the data that can identify key factors impacting the agency's facilities and mitigate their negative effects. **Requirements: R or Python, SQL, Data visualization (e.g. Power BI, Tableau, etc.), Forecasting techniques (Rolling Averages, SMA's, ARIMA, etc.…), Machine learning and Linear regression** **Internal and public data**
9	Electronics & IT Services	Lead scoring and insights	In this project, you will help a global electronics and IT services corporation optimize its list of new business prospects and improve its own benchmark models. To do this you will analyze several new data sources - including unstructured comments (i.e. an open text field) and use Streamlit, an open-source Python framework, to visualize your insights to help the organization leverage the model across its various businesses. **Requirements: Python (e.g. Streamlit), NLP, Data visualization** **Internal data**
10	Electronics & IT Services	Lead value modeling	In this project, you will help a global electronics and IT services corporation identify the unique factors that contribute to it securing its largest deals. Analyzing historical data, you will cluster and segment companies to target, and determine the best engagement tactics. In addition, you will make your insights available through easy-to-use visualizations. You will also help the company identify the Machine Learning Operations (MLOps) best practices to maintain the models for future use. **Requirements: Python** **Internal data**

11	Electronics & IT Services	Upsell modeling and engagement strategy	In this project, you will help a global electronics and IT services corporation sell new products to current clients based on the products they already have. The company has a significant client base and is always looking for new ways to support its customers. Although some basic models have been completed, the company is looking for your help to create a new engagement strategy, with the opportunity to build a two-stage recommender system that includes web content interactions and answers the question, "Which products should we show to which groups of customers?" You will then identify best practices to operationalize your models on the company's website using the current tech stack. **Requirements: Python** **Internal data**
12	Financial Services	Building machine learning models to bolster Anti-Money Laundering (AML) efforts	In this Capstone project, you will build machine learning models to bolster Anti-Money Laundering (AML) efforts for a multinational financial services company. The machine learning model(s) you build will enhance the sponsor's AML efforts by identifying suspicious patterns using a wide variety of data points. **Requirements: Python, Machine learning, Desire to learn about Anti Financial Crime Modeling.** **Internal data**
13	Environment (Agriculture)	Predicting and visualizing the environmental impact of pesticides and crop protection chemicals	In this Capstone project, you will use machine learning and data visualization to predict and visualize the environmental impact of various pesticides and crop protection chemicals for a global agrochemical company. In the U.S., crop protection chemicals (i.e. pesticides and agrochemicals in general) are regulated by the Environmental Protection Agency (EPA). The EPA Pesticide in Water Calculator (PWC) is typically used to predict the estimated environmental concentrations (or EECs) of pesticide residues in aquatic systems. However, when the PWC is applied to a wide range of pesticide use patterns across different geographies, it can be a great challenge to screen

			the impact of changes without having to perform numerous and repetitive manual simulations. Your goal will be to create a machine learning-based “meta model” from the PWC outputs and integrate this into a R Shiny application Graphical User Interface (GUI) to enable automated, instant and multiparametric visualization and scoping for the sponsor's product safety assessments. **Requirements: R or Python, Machine learning, Data Visualization** **Internal and public data**

14	International Affairs	Developing criteria and validation models to test Artificial Intelligence (AI) tool safety	Artificial Intelligence (AI) tools are increasingly critical in sensitive areas such as misinformation detection, bias mitigation, and accuracy control. The increasing reliance on AI tools necessitates robust validation frameworks to ensure their accuracy and fairness. In this Capstone project, you will work with a large intergovernmental organization to develop a certification script and process that evaluates AI tool performance across the key metrics of misinformation detection, bias mitigation, and accuracy control. Using globally recognized frameworks, you will test and evaluate your models, defining specific, measurable criteria for a “pass” in each evaluation area. Additionally, you will deliver detailed evaluation reports that provide insights into AI tool fairness, reliability, and transparency. The work you do will contribute to ensuring AI tools are used to serve the good of the international community. **Requirements: R, Machine learning, Data visualization (e.g. Power BI / Tableau / Qlik)** **Preferred skills: Python, NLP, Experience with survey tool sets** **Public data**
15	Manufacturing (Machinery/Equipment)	Customer analysis to determine	In this project, you will analyze the account base of a leading distributor of manufacturing instruments. The sponsor operates in several different

		seasonality and boost sales	industries (e.g. Oil & Gas, Chemical & Petrochemical, and Food & Beverage), providing the infrastructure that makes its customers' processing and manufacturing run more smoothly. You will help the sponsor identify which customers are worth further attention and when is the best time to engage with them (i.e. seasonality). Using prior purchasing behavior, company size, and product similarity across product sets and within industries, you will create an algorithmic, repeatable process that can create a simplified set of recommendations that sales representatives can look up and use when developing further business or doing quotes with a customer. The tool you develop will generate additional sales and help the sponsor better meet its customers' needs. **Requirements: R or Python, SQL, Experience in or a desire to learn data filtering and reduction, extraction, and data categorization** **Internal data**
16	Environment (Water)	Forecasting wastewater quality based on incoming flow and weather conditions	In this Capstone project, you will work with a global environmental services company to forecast wastewater quality based on the incoming flow of wastewater ("influent") and weather conditions. Weather significantly impacts the influent loads coming into the wastewater treatment plant, as well as the pollution concentration of wastewater. Plant operators must adjust treatment processes to adapt to these changes. The model you create will provide more time for plants to adjust system operations, thereby improving monitoring performance, optimizing wastewater treatment processes, and improving wastewater quality. **Requirements: R or Python, Machine learning, Predictive modeling** **Internal data**
17	Environment (Water)	Performing a big data analysis in a	In this Capstone project, you will work with a global environmental services company to more accurately predict process anomalies at

		wastewater treatment plants to predict future performance	wastewater treatment plants. Data from wastewater treatment plants is continuously collected and monitored, but this data is often underutilized. The goal of this project is to use data analytics and machine learning to produce predictions based on available data. These analytics can aid plant operators by providing more accurate predictions, and potentially warn operators in advance of impending process anomalies. The model you build will improve process control, detect process failures, identify inefficiencies, and continuously optimize performance. It will also help to reduce operational costs of wastewater treatment plants. **Requirements: R or Python, Machine learning, Predictive modeling** **Internal data**
18	Environment (Water)	Optimizing energy efficiency and cost savings for wastewater treatment plants	In this Capstone project, you will work with a global environmental services company to optimize efficiency and cost savings for wastewater treatment plants. Aeration is the process of adding oxygen to wastewater, and it can consume a large portion of a plant's energy cost. Aeration control with ammonia is crucial in wastewater treatment. Adjusting aeration and monitoring ammonia levels can reduce energy consumption significantly. Your goal is to create a self-learning algorithm that can generate predictions for the amount of ammonia needed in the influent wastewater up to 48 hours in advance. This prediction will help to determine the amount of aeration required for the plant, thereby improving energy savings and making existing operations more efficient. **Requirements: R or Python, Machine learning, Predictive modeling** **Internal data**
19	Manufacturing (Motor Vehicles)	Customer and competitor analysis for heavy-duty truck segment	In this Capstone project, you will work with a multinational heavy-duty truck manufacturer to analyze customer data to gain insights into current customers and strategically identify potential customers for conversion. You will develop methods for analyzing, integrating, and harmonizing company data using AI and machine learning to automate a process that

			is currently manual. You will deliver a report, an AI/ML model, and a dynamic dashboard designed for practical use by the sponsor beyond the course. **Requirements: R or Python, Machine learning** **Internal data**

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